

Physical Chemistry (I)

Lecture 09

1차 중간고사 안내 및 복습



1차 중간고사 안내

- 시간: 4월 9일(화) 오후 1:30~2:30
- 장소: 화학관 315호(홀수 학번), XXX호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

- 1장 "열역학의 기본 개념들"
- 2장 "기체의 성질"
- 3장 "열역학 제1법칙"
- 4장 "엔트로피"

시험 문제

총 여섯 문제

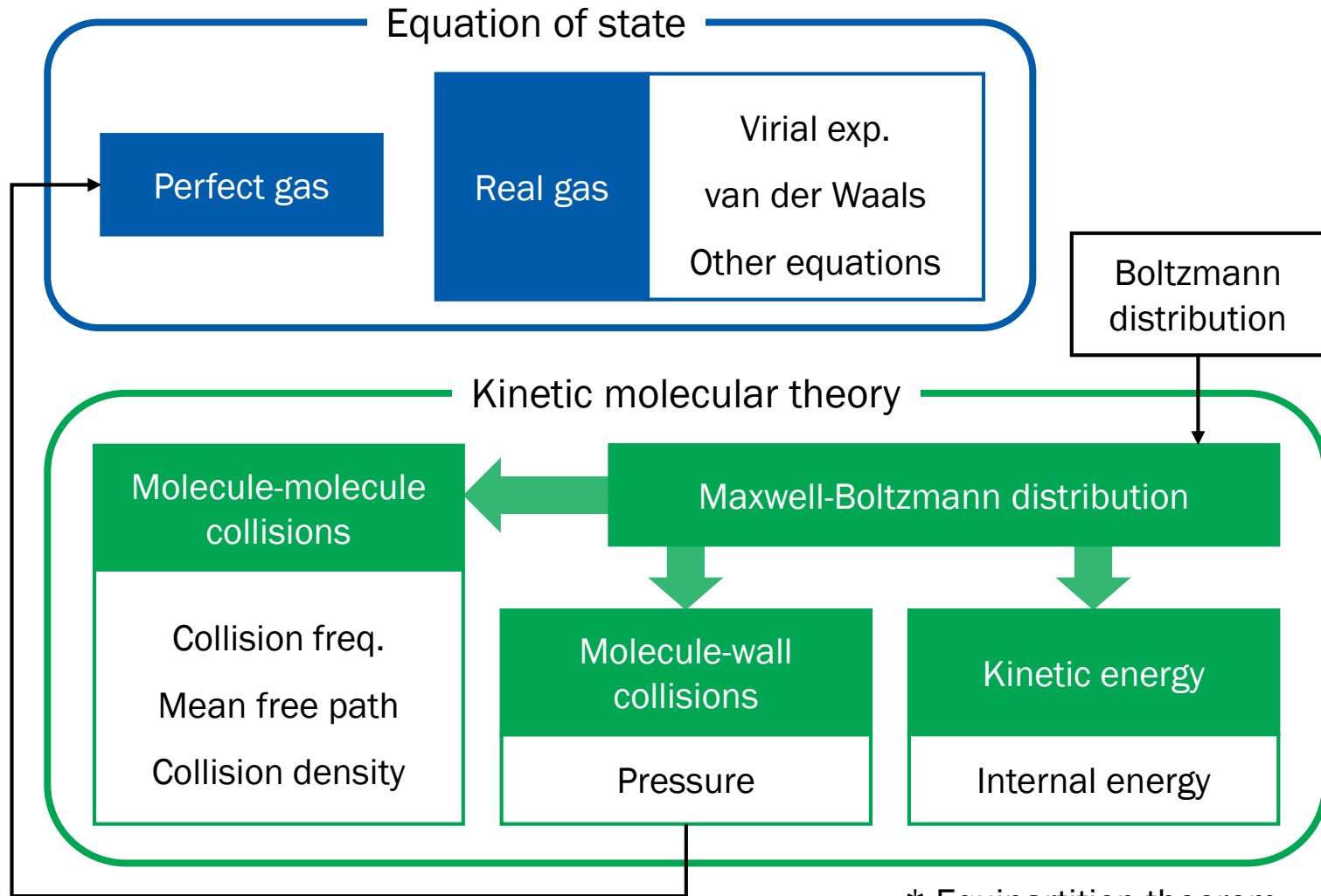
- 과제 문제 혹은 그 변형(3문제)
 - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(3문제)
 - 강의 자료 기준

Concepts in Chpt 1

- System and surroundings
- Classification of systems
- System properties: extensive vs. intensive
- Equilibria: mechanical, diffusive, and thermal
- The zeroth law of thermodynamics
- State of a system
- State change



Chapter 2



* Equipartition theorem

Concepts and Derivations in Chpt 2

- [C] Perfect gas vs. real gas
- [C] Equation of state, perfect gas law
- [C] Kinetic molecular theory
- [C] Boltzmann distribution
- [D] Maxwell-Boltzmann distribution
- [D] Characteristic speeds
- [D] Collision frequency, mean free path, collision density
- [D] Pressure



Concepts and Derivations in Chpt 2

- [D] Internal energy
- [C] Equipartition theorem
- [C] Compression factor
- [C] Virial expansion
- [C] Boyle temperature
- [C, D] van der Waals equation
- [D] van der Waals and virial
- [C] Other real gas equations: no need to memorize



적분표의 활용

$$\int_0^{\infty} x e^{-kx^2} dx = \frac{1}{2k}$$

$$\int_0^{\infty} v e^{-Mv^2/2RT} dv = ?$$

$$\int_{-\infty}^{\infty} v e^{-Mv^2/2RT} dv = ?$$

Chapter 3 (1)

First Law of Thermodynamics

$$dU = \delta q + \delta w$$

Work

$$\delta w = -p_{\text{ex}}dV + \delta w_{\text{add}}$$

- Constant- p change
- Reversible change

Heat

$$\delta q = C dT$$

- Constant- V change

$$C_V = (\partial U / \partial T)_V$$

Thermodynamics

Mathematics

State function

Exact differential

Path function

Inexact differential

Differential form of U

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

- Internal pressure π_T
- Response functions α , κ_T

Chapter 3 (2)

Thermodynamics	Mathematics
State function	Exact differential
Path function	Inexact differential

Differential form of U

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

Enthalpy

$$H = U + pV$$

$$dH = \delta q_p$$

Differential form of H

$$dH = \left(\frac{\partial H}{\partial p} \right)_T dp + \left(\frac{\partial H}{\partial T} \right)_p dT$$

$$C_V = (\partial U / \partial T)_V$$

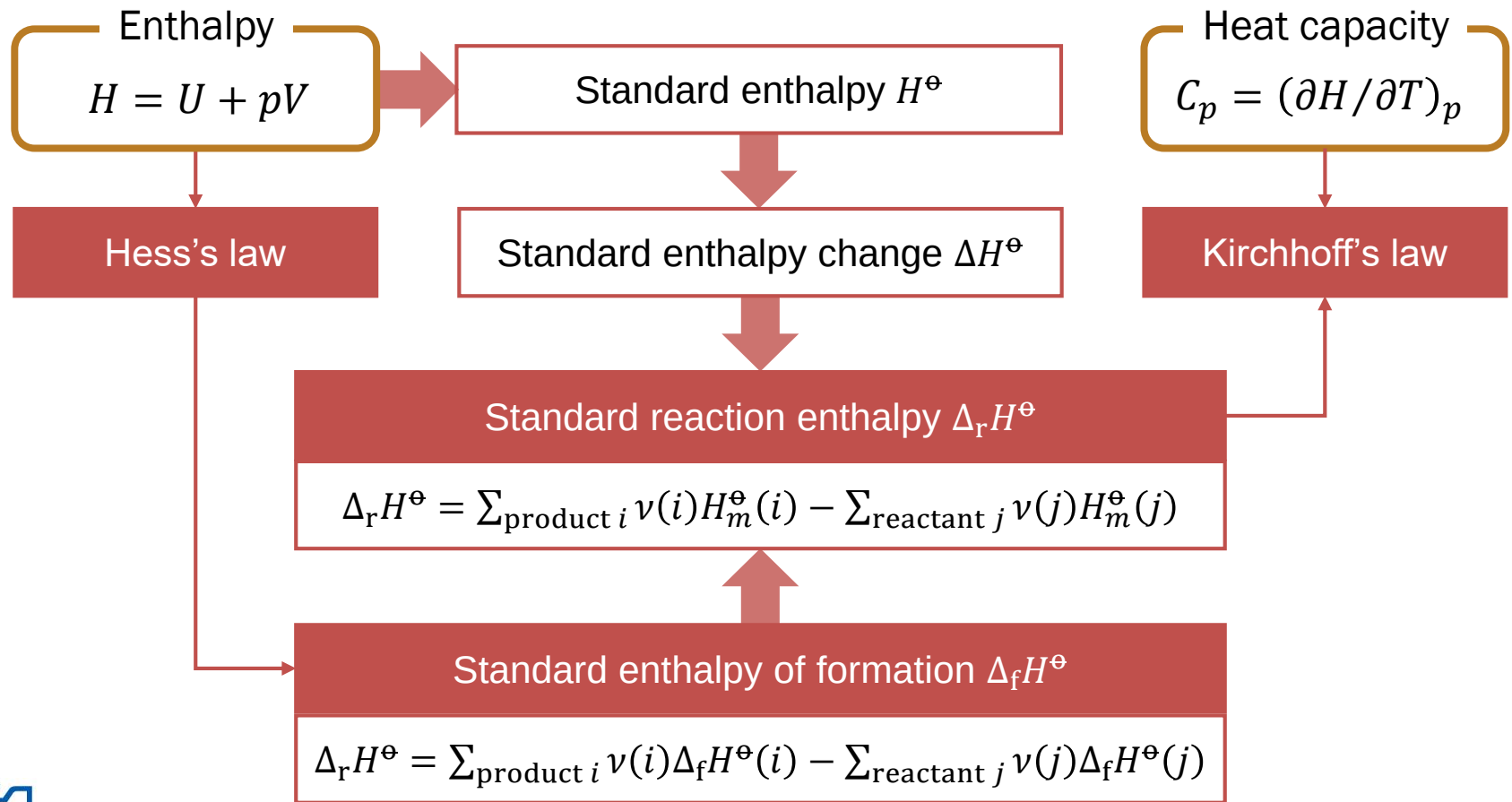
C_V and C_p

- Perfect gas
- Adiabatic change: $\delta q = 0$

$$C_p = (\partial H / \partial T)_p$$



Chapter 3 (3)



Concepts and Derivations in Chpt 3

- [C] First law of thermodynamics
- [D] Expansion work
 - Isochoric and isobaric change
 - Free expansion
 - Reversible (perfect gas, isothermal) change
- [D] Heat
 - Isochoric and isobaric change
 - Internal energy and isochoric heat capacity

Concepts and Derivations in Chpt 3

- [C] State and path functions
- [C, D] Differential form of internal energy
- [C] Response functions
- [D] Derivation of $(\partial U / \partial T)_p$
- [C, D] Enthalpy
 - Reversible isobaric change
 - Enthalpy and isobaric heat capacity
- [C, D] Differential form of enthalpy



Concepts and Derivations in Chpt 3

- [D] Heat capacities for perfect gas
- [C, D] Adiabatic change
 - Work for perfect gas
 - Reversible, perfect-gas relations: V - T and V - p
- [C] Standard enthalpy
 - Standard reaction enthalpy
 - Standard enthalpy of formation
- [C] Hess's law
- [D] Kirchhoff's law



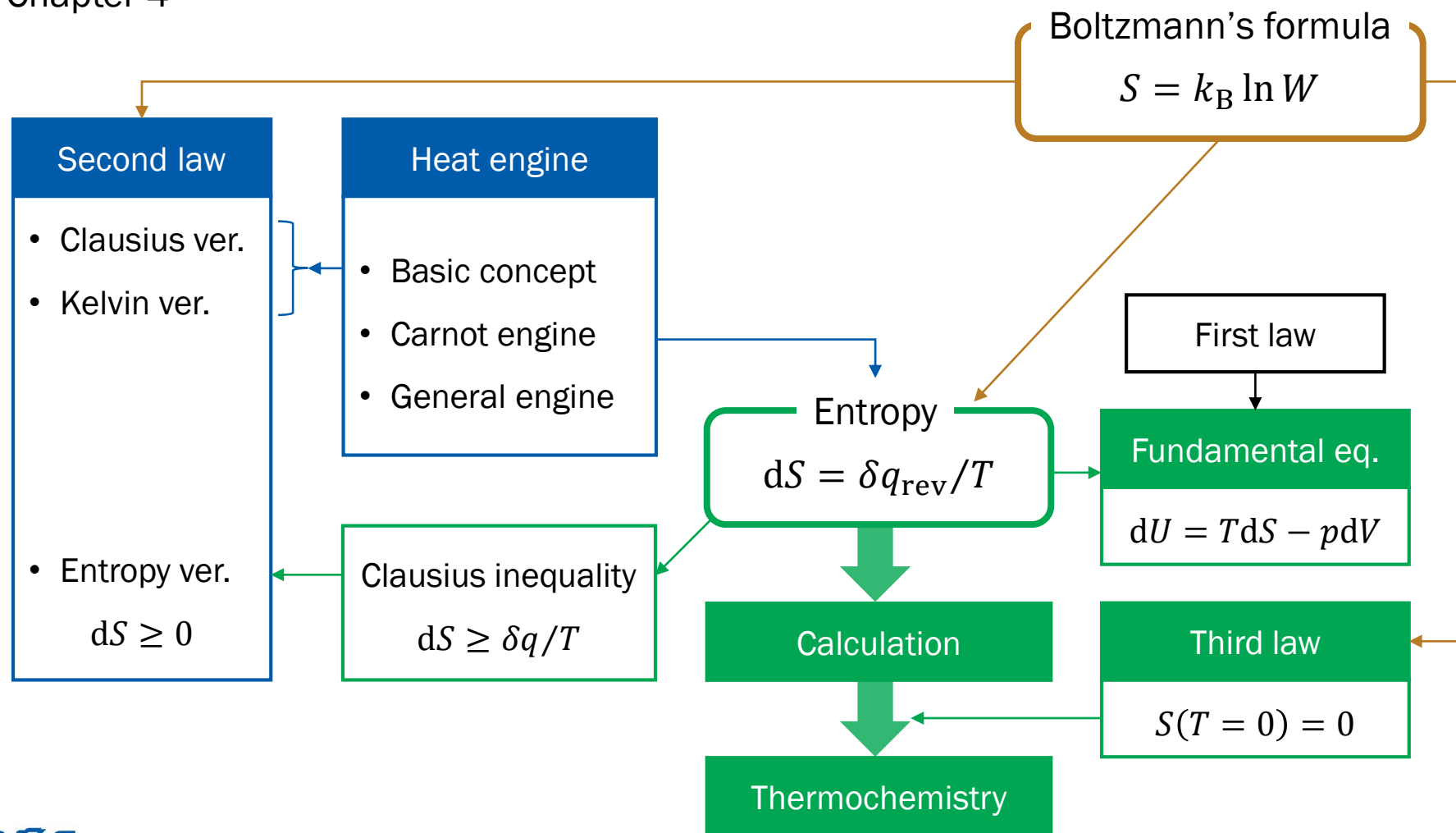
가역 변화

	$w = -\int p \, dV$	q	ΔU
등압($dp = 0$)	$-p\Delta V$		
등적($dV = 0$)	0		
등온($dT = 0$)			0
단열($\delta q = 0$)		0	

완전 기체의 가역 변화

"열 용량 일정"	$w = -\int p \, dV$	q	$\Delta U = C_V \Delta T$
등압($dp = 0$)	$-p\Delta V$	$C_p \Delta T$ $= C_V \Delta T + p\Delta V$	$C_V \Delta T$
등적($dV = 0$)	0	$C_V \Delta T$	$C_V \Delta T$
등온($dT = 0$)	$-nRT \ln(V_f/V_i)$	$nRT \ln(V_f/V_i)$	0
단열($\delta q = 0$)	$C_V \Delta T$	0	$C_V \Delta T$

Chapter 4



Concepts and Derivations in Chpt 4

- [C] Heat engine
- [C, D] The second law of thermodynamics
- [C] Engine efficiency
- [C, D] Carnot cycle and Carnot efficiency
- [C] Generalization of Carnot efficiency
- [C] Entropy
- [C, D] Clausius inequality
 - Entropy version of the second law
 - Entropy and spontaneity



Concepts and Derivations in Chpt 4

- [C, D] Fundamental equation of thermodynamics
- [D] Entropy calculations
 - Heat transfer
 - Entropy change of surroundings
 - Expansion
 - Phase transition
 - Heating
- [C] Nernst heat theorem and the third law

Concepts and Derivations in Chpt 4

- [C] Thermochemistry of entropy
 - Standard entropy
 - Standard reaction entropy
- [C] Entropy as an index of exhaustion
- [C] Boltzmann's formula
- [D] Applications of Boltzmann's formula
- [C] Reinterpretations of the laws
- [C, D] Independent spin model

